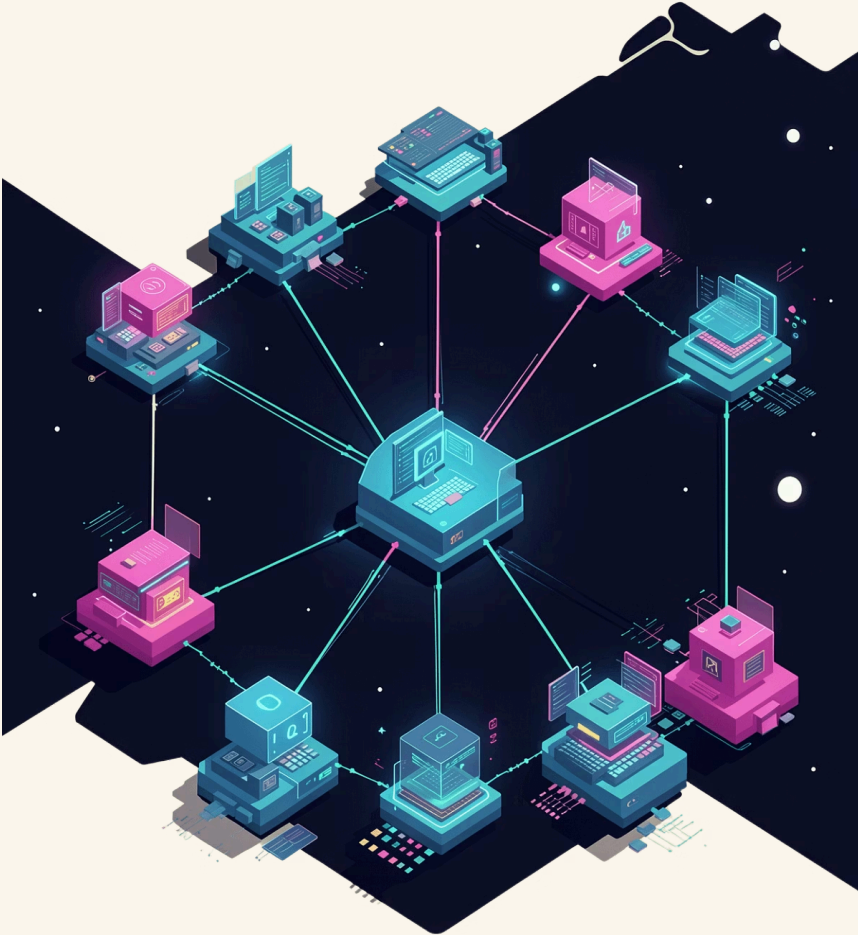




Multi-Agent Research Analyst

Moving from AI Agent to Agentic AI with LangGraph

Govardhan

The Problem

Shallow by Design

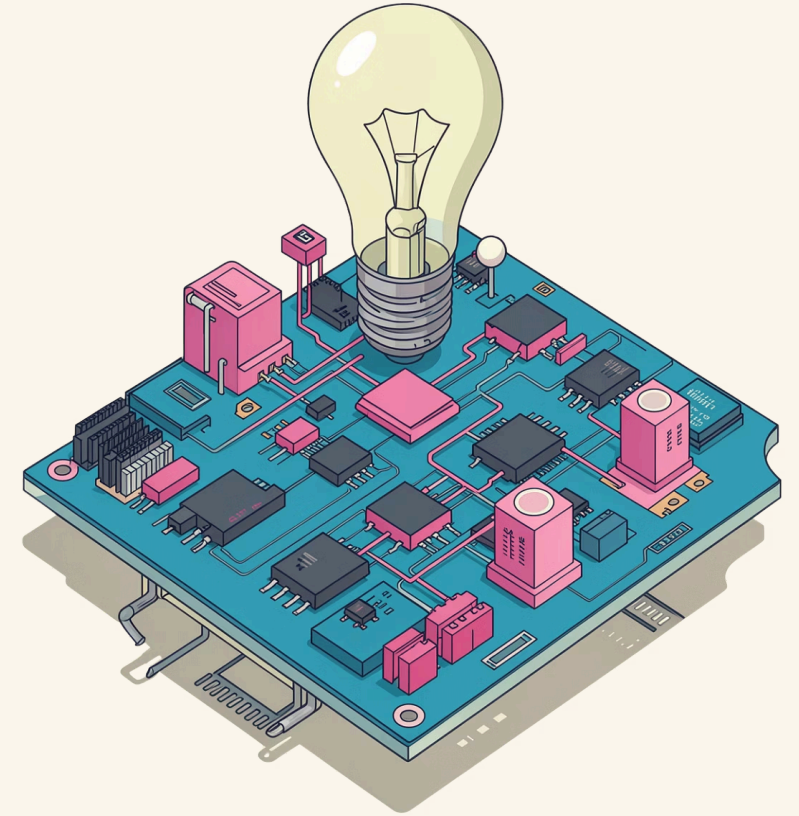
Single LLM calls are fast but shallow - they answer in one pass with no ability to verify or revisit.

Hallucinated Citations

Without grounding, models fabricate sources and cannot confirm whether claims are factually accurate.

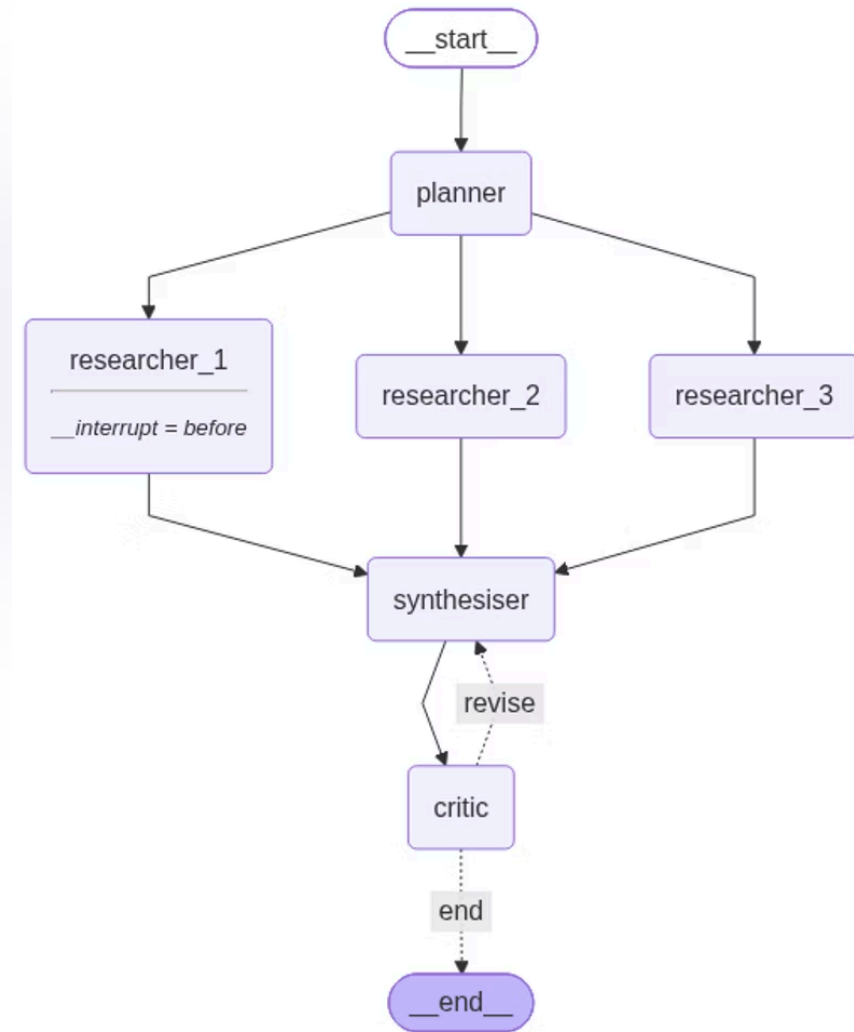
Complex Questions Need More

Real research requires planning, gathering evidence, and reviewing - not just a single prompt.



System Architecture

The LangGraph-powered pipeline coordinates a **Planner**, parallel **Researchers**, a **Synthesiser**, and a **Critic** - all sharing a typed state graph with an `operator.add` reducer to safely merge parallel outputs.



The 3 Agentic Patterns



Planner-Executor

The Planner decomposes the research question into focused sub-tasks before any execution begins.



Orchestrator-Workers

3 Researchers run in parallel. The `operator.add` reducer safely merges their outputs without overwriting.



Evaluator-Optimizer

The Critic scores the draft and loops it back to the Synthesiser until quality is approved. Max iterations: `MAX_ITERATIONS = 3`.

Tool Layer & State

Tools



Tavily Web Search

Live web retrieval for up-to-date sources.



Extract Page

Pulls full content from a target URL.



Safe Calculator

Evaluates numeric expressions safely.

Shared State

All agents communicate through a single **TypedDict** state object managed by LangGraph.

- `operator.add` reducer prevents parallel nodes from overwriting each other's data
- State persists across the full graph execution lifecycle
- `InMemorySaver` enables checkpointing and Human-in-the-Loop pauses

📄 The reducer is the key architectural decision that makes safe parallelism possible.

Evaluation Results

LLM-as-Judge scored both outputs 1–5 on **Coverage**, **Faithfulness**, **Citation Quality**, and **Coherence**.

```
=====
EVALUATING BASELINE VS MULTI-AGENT SYSTEM
=====

Evaluating: Compare the current EV strategies of Tata Motors and BYD....

-----
TOPIC                                BASE  MULTI  TIME B  TIME M  WINNER
-----
Compare the current EV strategies of Tata Mot...  2.8   4.8    7s     7s  Multi-Agent
-----
AVERAGES                                2.8   4.8    7s     7s

Score Improvement: +2.0 points
Time Trade-off: +-1 seconds average

Results saved to evaluation_results.json
```

TOPIC

Compare the current EV strategies of Tata Motors and BYD

2.75

Baseline Score

Single-agent output

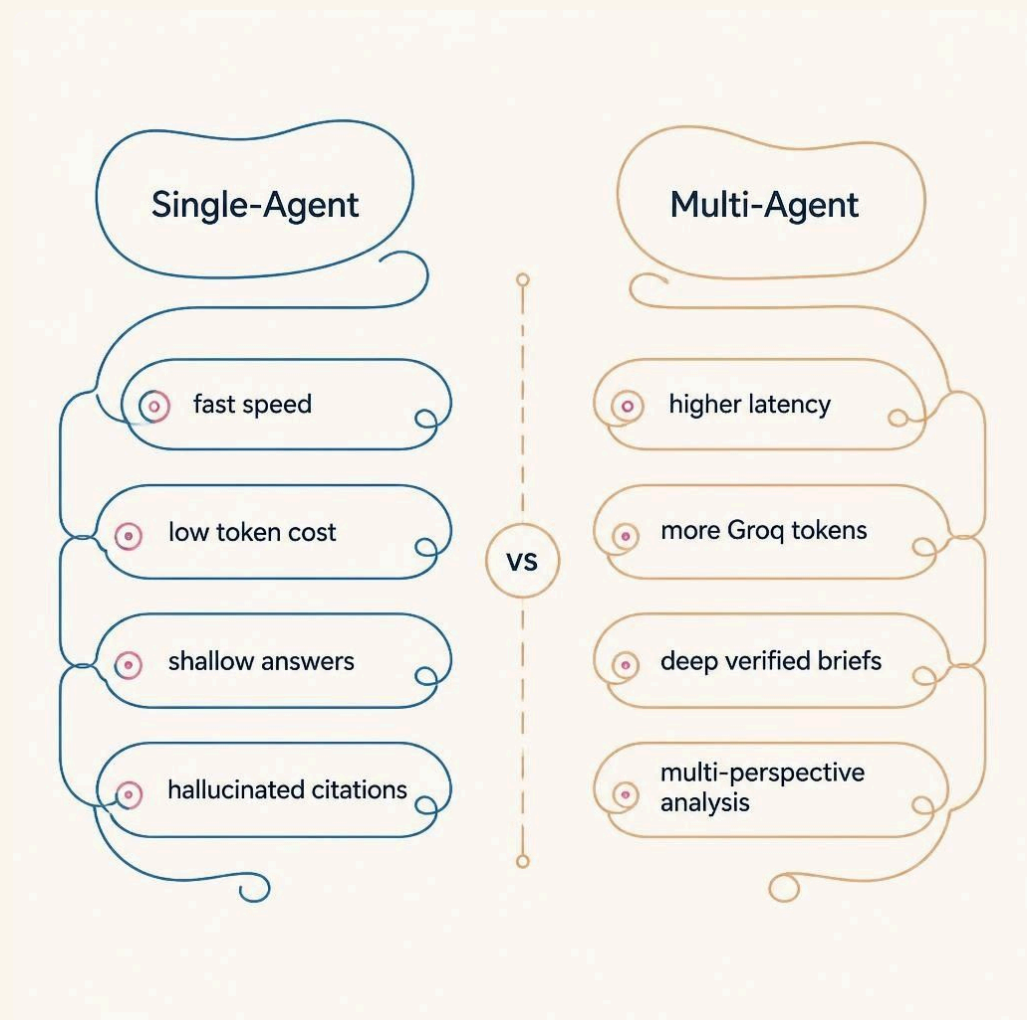
4.75

Multi-Agent Score

LangGraph pipeline

- Baseline time: **7.37 seconds**
- Multi-agent time: **6.61 seconds**
- Winner: **Multi-Agent**

Cost vs. Quality Trade-off



When is the trade-off worth it?

Agentic AI trades higher cost and latency for higher reliability. The single-agent baseline is fast and cheap, but frequently produces shallow answers or hallucinates citations.

- Multi-agent consumes significantly more **Groq tokens** across planner, researcher, and critic calls
- Yields structured, **multi-perspective briefs** with verified citations
- Trade-off is worth it for **high-stakes analysis**

✓ For enterprise research tasks, reliability outweighs raw speed.

Real-World Bugs & Mitigations

1

Groq/Llama 3.3 XML Hallucination

Bug: Native Tool Calling caused the model to hallucinate malformed XML tool calls, breaking the agent loop.

Fix: Built a custom text-parsing ReAct loop to bypass Native Tool Calling entirely and parse outputs manually.

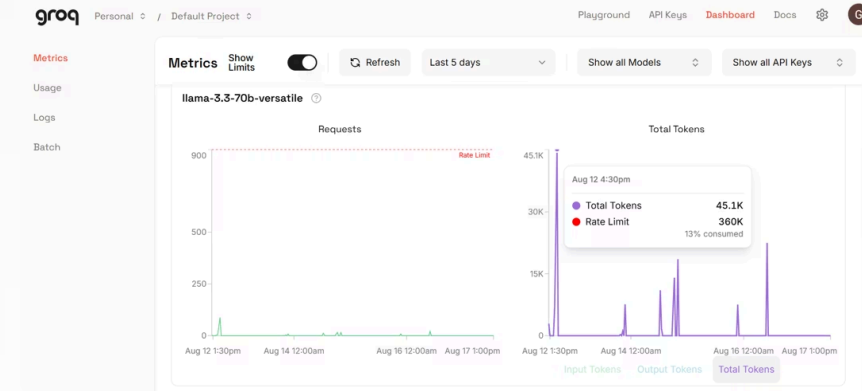
2

HTTP 429 Rate Limiting

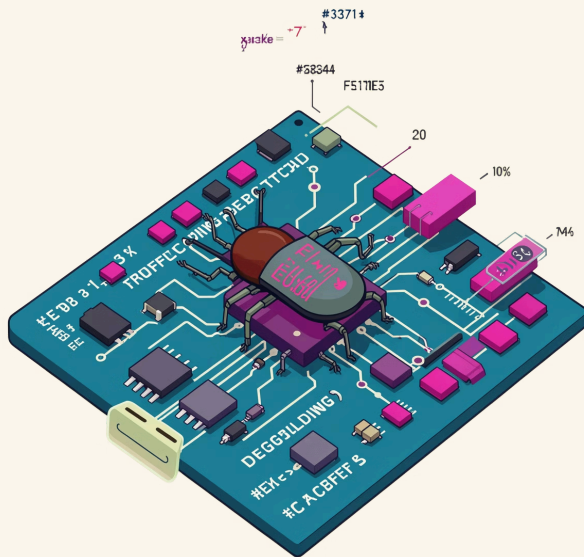
Bug: Parallel researcher execution on the Groq free tier triggered rate-limit errors, halting the pipeline.

Fix: Exponential backoff retry logic — `time.sleep(10 * attempt)` — with graceful degradation on repeated failures.

Note: These bugs highlight the infrastructural requirements of Agentic systems - error handling is as critical as model selection.



Advanced Features



Persistence via InMemorySaver

Graph state is checkpointed at every node, enabling resumable workflows and mid-run inspection.



Human-in-the-Loop

The graph pauses after planning to allow a human to approve or edit the research plan before execution continues.



Live Streamlit Deployment

The full multi-agent pipeline is deployed as an interactive web app built with **Streamlit**.

Conclusion



A Data-Engineering Challenge

Agentic AI is not just prompt engineering — it demands rigorous state management, reducer design, and fault-tolerant tooling.



State & Error Handling Win

State management and error handling matter more than prompt tuning. The `operator.add` reducer and exponential backoff were the real breakthroughs.



The Future of Enterprise AI

Multi-agent systems - built on frameworks like **LangGraph** with models like **Groq Llama 3.3 70B** - are the future of reliable, verifiable enterprise AI.

